

Quantum Computing Links

This document favors official documentation, educational resources, primary company pages, and widely used open-source projects. Each title is a clickable hyperlink; plain URLs are included for portability.

Sections

1. Basic intro and nontechnical explainers
2. Algorithms and theory
3. Programming frameworks and SDKs
4. Cloud platforms and managed access
5. Company and hardware ecosystem pages
6. Research, benchmarks, and communities

1. Basic intro and nontechnical explainers

Resource	Why it is useful	URL
IBM Quantum Learning	Structured learning path for quantum mechanics, circuits, and introductory quantum computing.	https://quantum.cloud.ibm.com/learning/en
Microsoft Quantum learning hub	Self-paced tutorials and conceptual introductions from Microsoft Quantum.	https://quantum.microsoft.com/learn
Google Quantum AI educational resources	Beginner-friendly resources and interactive material from Google Quantum AI.	https://quantumai.google/resources
Qiskit textbook / learning materials	Open learning materials covering quantum information, circuits, and algorithms.	https://github.com/qiskit-community/qiskit-textbook
Quantum Country	Essay-style introduction using spaced repetition to build intuition.	https://quantum.country/
Quantum Computing for the Very Curious	Accessible interactive essay by Andy Matuschak and Michael Nielsen.	https://quantum.country/qcvc
Quantum computing explained - IBM	High-level overview of concepts, hardware, and use cases.	https://www.ibm.com/topics/quantum-computing

2. Algorithms and theory

Resource	Why it is useful	URL
Qiskit learning: quantum algorithms	Course-style introduction to core quantum algorithms.	https://quantum.cloud.ibm.com/learning/en/courses/fundamentals-of-quantum-algorithms
Grover's algorithm - Qiskit textbook	Hands-on notebook for amplitude amplification and unstructured search.	https://github.com/Qiskit/textbook/blob/main/notebooks/ch-algorithms/grover.ipynb

Quantum phase estimation - Qiskit textbook	Notebook explaining phase kickback, inverse QFT, and phase estimation.	https://github.com/qiskit-community/qiskit-textbook/blob/main/content/ch-algorithms/quantum-phase-estimation.ipynb
Shor's algorithm - Qiskit textbook	Hands-on exploration of period finding and integer factorization.	https://github.com/qiskit-community/qiskit-textbook/blob/main/content/ch-algorithms/shor.ipynb
Quantum Approximate Optimization Algorithm - Qiskit	Introduction to QAOA and variational optimization.	https://learning.quantum.ibm.com/course/variational-algorithm-design/quantum-approximate-optimization-algorithm
Variational Quantum Eigensolver - Qiskit	VQE and related variational algorithms for chemistry and optimization.	https://learning.quantum.ibm.com/course/variational-algorithm-design/instances-and-extensions
Quantum algorithms: an overview - Ashley Montanaro	Survey paper covering applications including cryptography, search, optimization, simulation, and linear systems.	https://arxiv.org/abs/1511.04206
Quantum Algorithm Zoo	Catalog of quantum algorithms with references and complexity notes.	https://quantumalgorithmzoo.org/

3. Programming frameworks and SDKs

Resource	Why it is useful	URL
Qiskit documentation	IBM's open-source SDK documentation for circuits, transpilation, primitives, and runtime workflows.	https://docs.quantum.ibm.com/
Qiskit GitHub	Source code and issues for the Qiskit SDK.	https://github.com/Qiskit/qiskit
Cirq - Google Quantum AI	Python framework for writing, manipulating, optimizing, and simulating quantum circuits.	https://quantumai.google/cirq
Cirq documentation	Getting-started guides and API documentation for Cirq.	https://quantumai.google/cirq/start/start
PennyLane	Python framework for quantum machine learning, optimization, chemistry, and hybrid quantum-classical workflows.	https://pennylane.ai/
PennyLane documentation	Tutorials and API docs for differentiable quantum programming.	https://docs.pennylane.ai/
Microsoft Quantum Development Kit / Q# docs	Documentation for Q#, Python workflows, and Azure Quantum job submission.	https://learn.microsoft.com/en-us/azure/quantum/
Amazon Braket developer guide	AWS documentation for building and running quantum programs through Braket.	https://docs.aws.amazon.com/braket/latest/developerguide/what-is-braket.html
ProjectQ	Open-source framework for quantum computing in Python.	https://projectq.ch/
QuTiP	Python package for simulating open quantum systems and quantum dynamics.	https://qutip.org/

4. Cloud platforms and managed access

Resource	Why it is useful	URL
IBM Quantum Platform	Cloud access to IBM quantum systems, learning resources, and workloads.	https://quantum.cloud.ibm.com/
Azure Quantum	Microsoft's quantum cloud service and ecosystem.	https://azure.microsoft.com/en-us/products/quantum
Amazon Braket	AWS managed service for quantum hardware, simulators, and hybrid algorithms.	https://aws.amazon.com/braket/
Google Quantum AI	Google's quantum research, software, and experimental resources.	https://quantumai.google/
D-Wave Leap	Cloud access to D-Wave quantum annealing and hybrid solvers.	https://cloud.dwavesys.com/leap/

5. Company and hardware ecosystem pages

Resource	Why it is useful	URL
IBM Quantum	IBM's quantum hardware, software, roadmap, and services.	https://www.ibm.com/quantum
Google Quantum AI	Google's quantum hardware and software research group.	https://quantumai.google/
Microsoft Quantum	Microsoft's quantum computing research, tools, and solutions.	https://quantum.microsoft.com/
IonQ	Trapped-ion quantum computing company and hardware provider.	https://ionq.com/
Quantinuum	Integrated quantum computing company focused on hardware, software, and applications.	https://www.quantinuum.com/
Rigetti Computing	Superconducting quantum computing systems and cloud services.	https://www.rigetti.com/
D-Wave Quantum	Quantum annealing and hybrid quantum optimization systems.	https://www.dwavesys.com/
PsiQuantum	Photonic quantum computing company.	https://www.psiquantum.com/
Pasqal	Neutral-atom quantum computing company.	https://www.pasqal.com/
QuEra Computing	Neutral-atom quantum computers and cloud-accessible systems.	https://www.quera.com/

6. Research, benchmarks, and communities

Resource	Why it is useful	URL
arXiv quantum physics	Preprint archive for quantum information, quantum algorithms, and related physics.	https://arxiv.org/archive/quant-ph
Nature Quantum Information	Open-access journal for quantum information science.	https://www.nature.com/npjqi/
Quantum journal	Open-access journal for quantum science and technology.	https://quantum-journal.org/
Quantum Open Source Foundation	Community organization supporting open-source quantum software.	https://qosf.org/
Unitary Fund	Nonprofit supporting open quantum technology and ecosystem projects.	https://unitary.fund/
Quantum Economic Development Consortium	Industry consortium focused on quantum ecosystem development.	https://quantumconsortium.org/
NIST Quantum Information Program	U.S. standards and research resources related to quantum information science.	https://www.nist.gov/quantum-information-program

NSF Quantum Leap Challenge Institutes	U.S. National Science Foundation quantum research institute program.	https://new.nsf.gov/funding/initiatives/quantum-leap-challenge-institutes
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